

Skills Summary

Powers of Ten and Place Value

Use this reference while completing your worksheet or when studying.

Skill 1 — Multiplying by 10, 100, or 1,000

Rule: multiplying by a power of ten shifts every digit **left** — one place per factor of ten. In practice: attach that many zeros.

Example: $8 \times 1,000$

Step 1. Three factors of ten, so shift the 8 three places left (attach three zeros).

Step 2. $8 \times 1,000 = 8,000 \Rightarrow$ **8,000**

Try it: $32 \times 100 = ?$

check: 3,200

Skill 2 — Dividing by 10, 100, or 1,000

Rule: dividing by a power of ten shifts every digit **right** — one place per factor of ten. In practice: remove that many zeros.

Example: $4,700 \div 100$

Step 1. Two factors of ten, so shift the digits two places right (remove two zeros).

Step 2. $4,700 \div 100 = 47 \Rightarrow$ **47**

Try it: $9,000 \div 10 = ?$

check: 900

Skill 3 — Exponent Form

Rule: 10^n means n factors of ten — a 1 followed by n zeros. The exponent counts the zeros, and it counts the place shifts too.

Example: 6×10^3

Step 1. The exponent 3 says: shift the 6 three places left.

Step 2. $6 \times 10^3 = 6 \times 1,000 = 6,000 \Rightarrow$ **6,000**

Try it: $10^4 = ?$

check: 10,000

Skill 4 — Place-Value Shifts

Rule: the same digit one place to the **left** is worth **10 times** as much; one place to the **right** is worth one tenth as much.

Example: The 9 in 92,000 compared with the 9 in 9,200.

Step 1. Find each value from its place: 90,000 (ten-thousands) and 9,000 (thousands), one place apart.

Step 2. $90,000 \div 9,000 = 10 \Rightarrow$ **10** times as much

Try it: The 2 in 2,500 is how many times the 2 in 250?

check: 10